

Homework 6G: TM and Decidability

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You should submit a typeset or *neatly* written pdf on Gradescope. The grading TA should not have to struggle to read what you've written; if your handwriting is hard to decipher, you will be asked to typeset your future assignments. Three bonus points if you use L^AT_EX, and our template. You may collaborate with other students, but any written work should be your own.

1. Prove that the set of total functions from $\mathbb{N} \rightarrow \mathbb{N}$ is uncountable by diagonalization.

We will assume for contradiction that the set of all total functions from $\mathbb{N} \rightarrow \mathbb{N}$ is countable. Therefore, we are able to list all functions f as $f_0, f_1, f_2, \dots$. We wish to construct a function g such that $g(n) = f_n(n) + 1$. We trivially observe $g(n) \neq f_n(n)$. Therefore, since g cannot be equal to its corresponding f_n , we find that g is the function from $\mathbb{N}$ to $\mathbb{N}$ that does not appear in the enumeration. Contradiction! This contradiction implies our initial assumption of countability is false. Therefore, we find the set of total functions from $\mathbb{N}$ to $\mathbb{N}$ is uncountable. We find this proof to be similar to the proof claiming that "if S is an infinite countable set, then the powerset 2^S of S is uncountable".

2. Let

$$A_{CFG} = \{ \langle G, w \rangle \mid G \text{ is a CFG that generates string } w \}.$$

Prove that A_{CFG} is a decidable language.

We begin by recognizing that if w is in L_G , then there exists a derivation of w whose length is bounded by a function of $|w|$. Therefore, we find that only finitely many derivations must be examined. We construct a decider that enumerates all possible derivations of G up to the bound length and checks if any of them are able to produce w . If a derivation yielding w is found, the decider accepts. If no derivation yielding w is found, the decider rejects. Since we observe the number of derivations to verify is finite which can be done in finite time, the procedure ultimately halts. Therefore, since we have constructed a halting procedure for A_{CFG} , we find that A_{CFG} is decidable.

3. Consider a Language A such that A is undecidable but Turing acceptable (such as A_{TM}) show that $\overline{A}$ is non-Turing Acceptable.

We will assume for contradiction that A and $\overline{A}$ are Turing Acceptable. In layman's terms, we wish to prove there exists turing machines M_A and $M_{\overline{A}}$ that recognize A or $\overline{A}$, respectively. With an input x , we can run both machines in parallel until one of them accepts. We find that if M_A accepts, then $x \in A$. If $M_{\overline{A}}$ accepts, then $x \in \overline{A}$. This procedure would decide A which contradicts the initial assumption that A is undecidable. Therefore, we find that the assumption that $\overline{A}$ is acceptable is false. Therefore, we find $\overline{A}$ is non-Turing acceptable.

4. Prove $A_3 = \{\langle M, w \rangle \mid |L(M)| = 5\}$, the problem of deciding whether a machine accepts exactly 5 strings, is undecidable. (Use reduction from $HALT_{TM}$ to the problem in question.)

We will assume for contradiction there exists a decider for A_3 . Given an instance $\langle M, w \rangle$ of the halting problem, we are able to construct a new Turing Machine M' . On any input, we define that M' will simulate M on w . If the simulation halts, then M' checks if x is one of the five fixed strings. If it's determined x is one of these strings, then M' accepts. Otherwise, M' rejects. We find that if M does not halt on w , then M' will not halt on any input, which indicates it accepts no strings.

Therefore, we find if M halts on w , then $L_{M'}$ is the set $\{s_1, \dots, s_5\}$ and if M does not halt on w , then $L_{M'}$ is empty. Therefore, if we can decide if a given L has exactly five strings, then we can decide whether M halts on w by checking if $|L_{M'}| = 5$. Since the halting problem is undecidable, we find the assumption that there exists a decider for A_3 is false. Therefore, we find A_3 is undecidable.